Preview: set-01/by_level/Grade_03-04_Ecolier/y2025_q10 (v4)

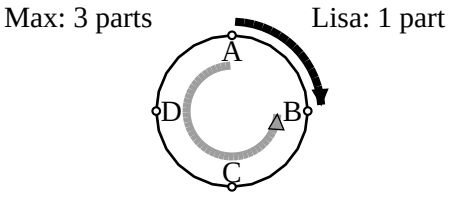
Running towards each other around a circle: between two meetings they cover one whole lap together

How to think about it: two runners going opposite ways round a circle

In these questions two children start at the same spot and run around a circle in **opposite directions**, so they keep meeting. You are asked where they meet, how often, or how many times one of them runs around.

The big idea: between two meetings, the two runners **together** run exactly **once around the circle**.

Why? Right after a meeting they stand on the same spot. They run apart, and they meet again when the gap between them is closed — that is when Lisa's piece and Max's piece together make the whole circle.



First meeting at B: 1 part + 3 parts = the whole circle

Step by step: the contest question

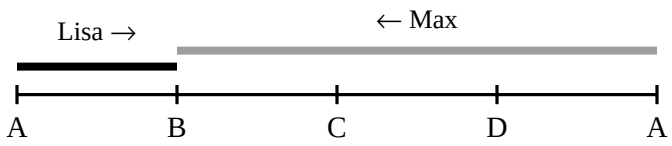
- **1. Cut the circle into equal parts.** The letters A, B, C, D cut it into 4 equal parts.
- **2. Who runs how much until the first meeting?** They meet at B. Lisa ran A to B: **1 part**. Max ran A to D to C to B: **3 parts**. Together 4 parts = once around.
- **3. It is the same every time.** From one meeting to the next, Lisa runs 1 more part and Max 3 more parts.
- **4. Keep count in a table.**

Meeting	Where	Lisa has run	Max has run
1	B	1 part	3 parts
2	C	2 parts	6 parts
3	D	3 parts	9 parts
4	A	4 parts	12 parts

- **5. Change parts into laps.** Max ran 12 parts, and 4 parts make one lap: 12 ÷ 4 = **3 times around**.

Another way to see it

Cut the circle open at A and lay it flat. Lisa starts at the left end, Max at the right end. They meet where their two pieces touch — the two pieces always fill the whole line.



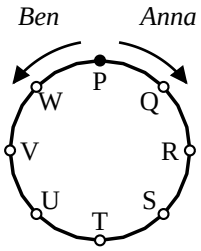
Watch out

- The number of **meetings** (4) is not the number of **laps** (3). Read what the question asks.
- Check your answer: Lisa's laps + Max's laps = number of meetings. Here 1 + 3 = 4.
- If they run equally fast, each runs half the circle: they meet exactly opposite the start.

Try it yourself: walk around a table with a friend or your brother, in opposite directions, one of you slowly and one fast. Stop every time you meet and count how many parts each of you walked.

Warm-up: small steps first (no points)

W1. Anna and Ben start together at point P and run around the circle in opposite directions. They run **equally fast**. At which point do they meet for the first time?

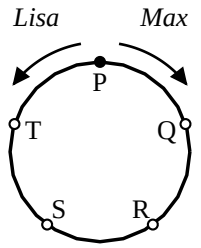


A. point P ***B.*** point Q ***C.*** point S ***D.*** point T ***E.*** point U

W2. A circle is cut into 8 equal parts. Zoe and Jack start together and run in opposite directions. When they meet for the first time, Zoe has run 3 parts. How many parts has Jack run?

A. 5 parts ***B.*** 6 parts ***C.*** 7 parts ***D.*** 8 parts ***E.*** 11 parts

W3. Max runs clockwise and Lisa anticlockwise (see picture). They start together at P and meet for the first time at Q. They keep running just as fast. Where do they meet the second time?



A. point P ***B.*** point Q ***C.*** point R ***D.*** point S ***E.*** point T

W4. A circle is cut into 4 equal parts. Every time Anna and Ben meet, Ben has run 3 more parts. How many parts has Ben run after 4 meetings?

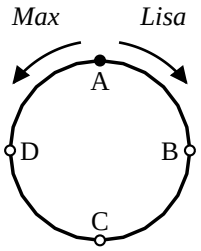
A. 12 parts ***B.*** 13 parts ***C.*** 14 parts ***D.*** 15 parts ***E.*** 16 parts

W5. A circle is cut into 6 equal parts. Tom runs 24 parts around the circle. How many times has Tom run around the circle?

A. 1 time ***B.*** 2 times ***C.*** 3 times ***D.*** 4 times ***E.*** 24 times

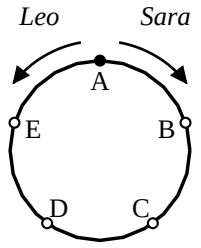
Part A: Each question is worth 3 points

1. Lisa and Max draw a big circle in the sand and write the letters A, B, C and D on it. They start at the same time from point A. Lisa runs clockwise and Max runs anticlockwise around the circle (see picture). They meet for the first time at point B, then at C, at D and then again at point A. How many times has Max run around the circle by then?



A. 1 time ***B.*** 3 times ***C.*** 4 times ***D.*** 5 times ***E.*** 8 times

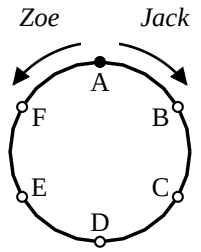
2. Sara and Leo draw a big circle in the sand and write the letters A, B, C, D and E on it. They start at the same time from point A. Sara runs clockwise and Leo runs anticlockwise around the circle (see picture). They meet for the first time at point E, then at D, at C, at B and then again at point A. How many times has Sara run around the circle by then?



A. 1 time ***B.*** 2 times ***C.*** 3 times ***D.*** 4 times ***E.*** 6 times

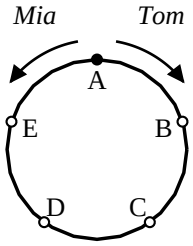
Part B: Each question is worth 4 points

3. Jack and Zoe start at the same time from point A of the circle. Jack runs clockwise, Zoe runs anticlockwise (see picture). They meet for the first time at point E, then at C and then again at point A. How many times has Jack run around the circle by then?



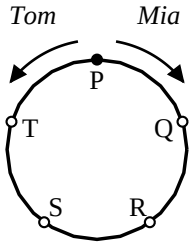
A. 2 times ***B.*** 3 times ***C.*** 4 times ***D.*** 5 times ***E.*** 6 times

4. Tom and Mia start at the same time from point A of the circle. Tom runs clockwise, Mia runs anticlockwise (see picture). They meet for the first time at point D, then at B, at E, at C and then again at point A. How many times has Tom run around the circle by then?



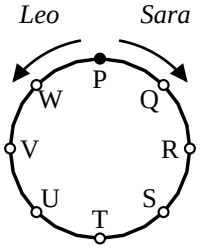
A. 1 time ***B.*** 2 times ***C.*** 3 times ***D.*** 4 times ***E.*** 5 times

5. Mia and Tom start together at point P and run around the circle in opposite directions: Mia clockwise and Tom anticlockwise (see picture). The points are evenly spaced. Mia runs 4 times as fast as Tom. At which point do they meet for the first time?



A. point P ***B.*** point Q ***C.*** point R ***D.*** point S ***E.*** point T

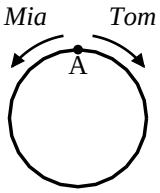
6. Sara and Leo start together at point P and run around the circle in opposite directions: Sara clockwise and Leo anticlockwise (see picture). The points are evenly spaced. Leo runs 7 times as fast as Sara. At which point do they meet for the first time?



A. point P ***B.*** point Q ***C.*** point S ***D.*** point U ***E.*** point W

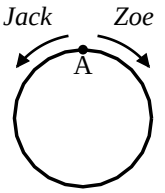
Part C: Each question is worth 5 points

7. Tom runs clockwise and Mia runs anticlockwise around a circle. They start together at point A. When they are together at point A again for the first time, Tom has run around the circle once and Mia 2 times. How many times did they meet? (Count the meeting at A, but not the start.)



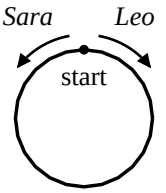
A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

8. Zoe runs clockwise and Jack runs anticlockwise around a circle. They start together at point A. When they are together at point A again for the first time, Jack has run around the circle 3 times and Zoe 5 times. How many times did they meet? (Count the meeting at A, but not the start.)



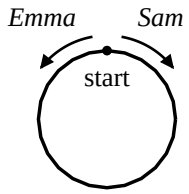
A. 2 ***B.*** 4 ***C.*** 5 ***D.*** 7 ***E.*** 8

9. Leo and Sara start together at the same spot of a round track. Leo runs clockwise and Sara runs anticlockwise. Sara needs 12 minutes to run once around, Leo only needs 4 minutes. After 12 minutes they stop — both are back at the start together. How many times did they meet? (Count the meeting at the end, but not the start.)



A. 1 ***B.*** 2 ***C.*** 3 ***D.*** 4 ***E.*** 5

10. Sam and Emma start together at the same spot of a round track. Sam runs clockwise and Emma runs anticlockwise. Emma needs 8 minutes to run once around, Sam also needs 8 minutes. After 24 minutes they stop — both are back at the start together. How many times did they meet? (Count the meeting at the end, but not the start.)



A. 1 ***B.*** 3 ***C.*** 5 ***D.*** 6 ***E.*** 7

Answer Key

W1.	**D**	**W2.**	**A**	**W3.**	**C**	**W4.**	**A**	**W5.**	**D**	**1.**	**B**
2.	**D**	**3.**	**A**	**4.**	**C**	**5.**	**E**	**6.**	**B**	**7.**	**C**
8.	**E**	**9.**	**D**	**10.**	**D**						

Solutions

W1. D (point T) — Equally fast means each runs half of the circle. Half of 8 gaps is 4 gaps: point T, right opposite P. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · w_equal_speed]

W2. A (5 parts) — Together they ran once around, that is 8 parts. 8 – 3 = 5. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · w_parts_first]

W3. C (point R) — Each time, Max runs 1 part (one point further) and Lisa the other 4 parts. So the second meeting is one point after Q: at R. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · w_next_meeting]

W4. A (12 parts) — 4 meetings × 3 parts = 12 parts. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · w_total_parts]

W5. D (4 times) — One time around is 6 parts. 24 ÷ 6 = 4. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · w_parts_to_laps]

1. B (3 times) — Between two meetings they run exactly once around the circle together. Lisa only gets to the next letter (1 of 4 parts), so Max runs the other 3 parts. 4 meetings × 3 parts = 12 parts = 3 times around.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · fast_laps]

2. D (4 times) — Between two meetings they run exactly once around the circle together. Leo only gets to the next letter (1 of 5 parts), so Sara runs the other 4 parts. 5 meetings × 4 parts = 20 parts = 4 times around.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · fast_laps]

3. A (2 times) — Each time, Zoe runs 2 of the 6 parts and Jack runs the other 4. They meet 3 times, so Jack runs 3 × 4 = 12 parts = 2 times around. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · skip_points]

4. C (3 times) — Each time, Mia runs 2 of the 5 parts and Tom runs the other 3. They meet 5 times, so Tom runs 5 × 3 = 15 parts = 3 times around. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · skip_points]

5. E (point T) — Until they meet they run once around the circle together, and Mia runs 4 times as far: Tom runs 1 part and Mia 4 parts of 5. 1 part of 5 is 1 of the 5 gaps, so Tom only gets to point T.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · where_first]

6. B (point Q) — Until they meet they run once around the circle together, and Leo runs 7 times as far: Sara runs 1 part and Leo 7 parts of 8. 1 part of 8 is 1 of the 8 gaps, so Sara only gets to point Q.
[set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · where_first]

7. C (3) — Between two meetings they run exactly once around the circle together. Together they ran 1 + 2 = 3 times around, so they met 3 times. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · count_meetings]

8. E (8) — Between two meetings they run exactly once around the circle together. Together they ran 3 + 5 = 8 times around, so they met 8 times. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · count_meetings]

9. D (4) — Between two meetings they run once around the track together. In 12 minutes Sara runs 1 and Leo runs 3 times around: 1 + 3 = 4 laps together, so 4 meetings. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · minutes]

10. D (6) — Between two meetings they run once around the track together. In 24 minutes Emma runs 3 and Sam runs 3 times around: 3 + 3 = 6 laps together, so 6 meetings. [set-01/by_level/Grade_03-04_Ecolier/2025 Q10 · minutes]